

approached the company, it did not know just what the STI was planning; but even if it did, very little would probably have changed.

Allerd and Moore were not looking to build a device that would take over the lives of the household – at least not just yet. The focus was on delivering good performance at an affordable price with a processor that could last a long time, but did not necessarily have to be a ‘mini-supercomputer’ like that of the PS3.

Using the PowerPC architecture, IBM developed the Xenon processor for the Xbox 360, which boasted of three cores with 32 KB of cache memory on each. Similar to the Cell Processor, the Xenon’s core clock speed was 3.2 GHz.

Microsoft was in a hurry to launch the device, though, and so ended up coming out with 90nm chips that took up more physical space, more power and emitted more heat.

Since its launch, the Xenon processor has undergone a change, and is now called the ‘Falcon’. In the Falcon, MS managed to bring the chips down to the 65 nm fabrication process, thus saving precious manufacturing costs and being able to scale down the size of the Xbox 360 if need be.

Xenos GPU

While Sony decided that Nvidia was the way to go for all things graphical, Microsoft was of the opinion that ATI would give more bang-for-buck.

The graphics processing unit (GPU) that ATI came out with was quickly dubbed the Xenos and promised a revolution for the Xbox 360. It was based on ATI’s upcoming R600 architecture, which was to be used for the company’s premium graphics card line-up.

To make the GPU less power-hungry, the company decided to merge the two major aspects of shader hardware: vertex and pixel. In a normal GPU, the shaders are on two physically different segments of the circuit; but in the Xenon, ATI put them in the same place.

As AnandTech puts it: “The unified shader architecture that ATI chose to use in their Xbox 360 GPU allows them to pack more functionality onto fewer transistors as less hardware needs to be duplicated for use in different parts of the chip and will run both vertex and shader programs on the same hardware.”

Still, this change in approach ended up taking a lot of ATI’s time and the resultant GPU was based on an 80nm fabrication process, thus throwing out more heat. The two 60 mm fans that accompanied the heatsink were never going to be enough – a fatal error that later led to the infamous ‘Red Ring Of Death’.

However, this aspect has since been corrected and the new version of the Xbox 360 based on the Falcon chipsets use a 65 nm fabrication process for the same.

The Xenos GPU has a core clock speed of 500 MHz and 10 MB of embedded RAM (eRAM).